PROJECT PROPOSAL (PROPOSED SOLUTION)

Project Initialization and Planning Phase

27 October 2026 | Team ID: LMP-EV-2026 | Yuvraj Shrivastava | EN25CS3011105

Date	27 October 2026
Team ID	LMP-EV-2026
Project	Electric Vehicle Market Performance & Insights
Project Lead / Sole Contributor	Yuvraj Shrivastava (EN25CS3011105), Medicaps University
Maximum Marks	3 Marks

1. Project Overview

Objective

To develop a public-facing, interactive Tableau analytics experience that helps users explore electric-vehicle market performance and technology patterns across vehicle categories, manufacturers, battery types, model years and manufacturing countries. The project presents key indicators and visual comparisons in one dashboard and makes the published analysis accessible through a GitHub Pages website.

Scope

The project covers preparation of the supplied EV training CSV, analysis in Tableau Desktop, creation of nine worksheets and an interactive dashboard, publication through Tableau Public, and embedding of the published view in a responsive HTML/CSS website hosted on GitHub Pages. The supplied test CSV is retained as a resource but is not connected to the Tableau workbook. The current Tableau Story has Story Point 1 completed; Story Points 2–4 are planned and are not represented as completed deliverables.

2. Problem Statement

Description

EV market information spans sales, vehicle segments, price, driving range, battery chemistry, customer ratings and manufacturing geography. Reviewing these dimensions separately across raw data or disconnected charts can make it difficult for a market analyst to compare performance and identify patterns. The project addresses this analysis and communication challenge by organizing the available dataset into a consolidated interactive dashboard.

Impact

The dashboard can reduce manual effort when exploring the supplied dataset, make comparisons easier through consistent visual views and filters, and provide a shareable starting point for discussion. Findings are descriptive of the dataset used and should not be treated as forecasts or as a complete representation of the entire global EV market.

3. Proposed Solution

Approach

• Use **ev_market_train.csv** as the connected Tableau data source; the separate test CSV is not connected.

• Prepare the training data by removing nulls and duplicate records and correcting data types.

• Build nine Tableau worksheets for executive KPIs, category sales, manufacturer share, battery performance/adoption, geographic footprint, price versus range, rating-related comparisons and a battery/category matrix.

• Combine the views into the EV Market Performance & Strategy Dashboard with filters for model year, category, battery type and manufacturing country, and enable relevant chart selections as filters.

• Publish the workbook to Tableau Public and embed the public visualization in the HTML/CSS project website hosted on GitHub Pages.

Key Features

• Executive KPI summary: annual sales units, distinct vehicle count, average range, average price in USD and average customer rating.

• Nine complementary visual worksheets for segment, manufacturer, battery, geography, price/range and rating-related analysis.

• Interactive dashboard filters and chart-selection actions to support exploratory analysis.

• Public Tableau visualization embedded in a responsive website, with direct access to Tableau Public.

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Resource Requirements and Delivery

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4. Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Development computer	One personal laptop/desktop. Exact CPU, GPU and RAM specifications were not provided. No dedicated GPU is required for this Tableau dashboard.
Computing Resources	Local data preparation, Tableau authoring and browser testing	CPU-based desktop/laptop workflow; no cloud compute allocation specified.
Memory	Runtime memory for Tableau Desktop and browser	Depends on the available computer; exact RAM not confirmed.
Storage	CSV files, Tableau workbook, website source and exported documentation	Local storage; exact capacity not confirmed.
Software	Analytics and visualization	Tableau Desktop for workbook authoring; Tableau Public for publication.
Frameworks	Website implementation	Static HTML and CSS; no Python web framework or application backend is used.
Libraries	Additional libraries	No separate analytical Python library dependency is claimed for this Tableau project.
Development Environment	Authoring and version control	Tableau Desktop, web browser, GitHub repository and GitHub Pages.
Data	Dataset source, size and format	SkillWallet-supplied EV market CSV resources. ev_market_train.csv is connected; ev_market_test.csv is supplied but not connected. CSV format; row count/file size not provided.

5. Ownership and Delivery

This is an individual project completed by Yuvraj Shrivastava alone. No other team member or contributor was involved. The website and Tableau workbook are public-facing deliverables. The project date recorded in this documentation is 27 October 2026.